

DETERMINATION OF DARCY'S COEFFICIENT OF FRICTION IN PIPES

AIM:-

1. To determine the friction factor 'f' of the given pipe (Darcy's constant)
2. To find the value of 'f' from the graph h_f Vs v^2 .
3. To determine the head loss due to friction factor for a flow rate ofcm³/s.

APPARATUS

1. A pipe provided with inlet and outlet valves.
2. Differential Mercury column 'U' tube manometer.
3. A collecting tank fitted with a piezometer.
4. Stop watch.

PRINCIPLE

The frictional factor 'f' is calculated by using the expression

$$\text{Darcy's equation, } h_f = \frac{4flv^2}{2gd}$$

Where

h_f = Loss of head due to friction = $(h_2 - h_1) \frac{(S_m - S_l)}{S_l}$ in cm.

$h_{1,}$ = Manometric head in left limb manometer in cm.

h_2 = manometric head in right limb manometer in cm.

S_m = Specific gravity of manometric liquid.

S_l = Specific gravity of flowing liquid.

f = Friction factor.

l = length of pipe between the pressure tapping cocks in cm.

v = velocity of flow in the pipe = $\frac{Q}{a}$ cm³/sec

OBSERVATIONS AND TABULATION							
Sl. No	Manometer readings		Time	Actual Discharge	Velocity of Flow		Coefficient of friction
	h_1	h_2	$h_f = (h_1 - h_2) / 12.6$	Q_a	V	V^2	f
Unit	cm	cm	cm of water	cm^3/s	cm/s	cm^2/s^2	
1							
2							
3							
4							
5							
6							

$$Q = \text{Actual discharge} = \frac{A \times R}{t} \quad \text{cm}^3/\text{sec}$$

A = Area of collecting in cm^2 .

R = height of water collected in cm.

t = Time for collection 'R' rise of water in s.

a = area of pipe in cm^2

g = Acceleration due to gravity.

$$\text{Friction factor, } f = \frac{h_f 2 g d}{4 l v^2}$$

PROCEDURE

1. The diameter of the pipe and area of collecting tank are measured.
2. Open the out let valve of the required pipe and close the bypass valve.
3. Start the pump and water is allowed to flow through the pipe.
4. The inlet valve of the manometer is opened and the manometric heads in both limbs ' h_1 ' and ' h_2 ' are noted.
5. Close the out let valve of the collecting tank and note the time 't' for 'R' cm rise by using a stop watch.
6. The above procedure is repeated by gradually decreasing the flow by partly open the bypass valve and observing the corresponding readings. (minimum 6 sets)
7. The observations are tabulated and the friction factor is calculated.

SAMPLE CALCULATIONS {SET No.....}

$$\text{Head loss between pressure tapings, } h_f = (h_2 - h_1) \frac{(S_m - S_l)}{S_l} \text{ in cm}$$

$$\text{Left limb manometer reading, } h_1 =$$

$$\text{Right limb manometer reading, } h_2 =$$

$$\text{Specific gravity of manometric liquid, } S_m = 13.6$$

$$\text{Specific gravity of flowing liquid, } S_l = 1$$

$$\text{Head loss between pressure tapings, } h_f = (\quad - \quad) \frac{(13.6 - 1)}{1} \text{ in cm} =$$

$$\text{Acceleration due to gravity, } g = 981 \frac{\text{cm}}{\text{s}^2}$$

Diameter of pipe, d = 1.6 cm
 Area of pipe, a = cm²
 Length of pipe, l = 100 cm
 Area of collecting, A = cm²
 Rise of water, R = 5 cm
 Time for collection, 'R' rise, t = s

Actual discharge, Q = $\frac{A \cdot R \text{ cm}^3}{t \text{ s}}$

= cm³/s

Cross section area of pipe, a = $\frac{\pi}{4} d^2$ = cm²

Velocity of flow, $v = \frac{Q}{a}$ = cm/s

Coefficient of friction, f = $\frac{h_f 2 g d}{4 l v^2}$ =

RESULT

Coefficient of friction, f =

Head loss, h_f =

INFERENCE

GRAPH

1. v^2 Vs h_f